

Lecture 9 Local Extrema Cont'd

Recap. Second derivative test

$f(x,y) \in C^2$, (x_0, y_0) is a critical point, i.e. $\nabla f(x_0, y_0) = \vec{0}$

$(f_{xx}f_{yy} - f_{xy}^2) _{(x_0, y_0)}$	$f_{xx}(x_0, y_0)$	(x_0, y_0) is a
> 0	> 0	local minimum
> 0	< 0	local maximum
< 0	anything	saddle point

Any other cases are inconclusive.

Pf. $f(x,y) = f(x_0, y_0) + \cancel{f_x(x_0, y_0)(x-x_0)} + \cancel{f_y(x_0, y_0)(y-y_0)}$
 $+ \frac{f_{xx}(x_0, y_0)}{2}(x-x_0)^2 + f_{xy}(x_0, y_0)(x-x_0)(y-y_0) + \frac{f_{yy}(x_0, y_0)}{2}(y-y_0)^2$
 $+ \text{higher-order terms}$

$A = f_{xx}(x_0, y_0)$, $B = f_{xy}(x_0, y_0)$, $C = f_{yy}(x_0, y_0)$
 $X = x - x_0$, $Y = y - y_0$

$f(x,y) \approx f(x_0, y_0) + \frac{1}{2}(AX^2 + 2BXY + CY^2)$

$AX^2 + 2BXY + CY^2 = \frac{1}{A}(A^2X^2 + 2ABXY + BY^2 - BY^2 + ACY^2)$

$= \frac{1}{A}[(AX + BY)^2 + (AC - B^2)Y^2]$

$(f_{xx}f_{yy} - f_{xy}^2) _{(x_0, y_0)}$	$f_{xx}(x_0, y_0)$	$AX^2 + 2BXY + CY^2$	near (x_0, y_0) , $f(x,y)$ is
$AC - B^2$			
> 0	> 0	always ≥ 0	always $\geq f(x_0, y_0)$ local minimum
> 0	< 0	always ≤ 0	always $\leq f(x_0, y_0)$ local maximum
< 0	anything	+ or -	$\geq f(x_0, y_0)$ or $\leq f(x_0, y_0)$ saddle point

Example. Let $f(x,y) = 3y^2 - 2y^3 - 3x^2 + bxy$. Find all critical points and determine the nature of each of them.

Sol. $\frac{\partial f}{\partial x} = -6x + by = 0$ $\frac{\partial f}{\partial y} = 6y - 6y^2 + bx = 0$

$$6x^4 = 2x \Rightarrow x^3 = \frac{1}{3}x \Rightarrow x = 0 \text{ or } x = 2$$

Two critical points: $(0, 0)$, $(2, 2)$

$$f_{xx} = -6, \quad f_{xy} = 6, \quad f_{yy} = 6 - 2y$$

Critical point	f_{xx}	f_{yy}	f_{xy}	$(f_{xx}f_{yy} - f_{xy}^2)$	nature
$(0, 0)$	-6	6	6	-72	saddle point
$(2, 2)$	-6	-18	6	72	local maximum

Example. (Inconclusive tests). a. $f(x, y) = 2x^4 + y^4$ b. $f(x, y) = 2 - xy^2$

Sol. a. $f_x = 8x^3$ $f_y = 4y^3 \Rightarrow (0, 0)$ is a critical point

$$(0, 0): \quad f_{xx} = 24x^2 = 0, \quad f_{xy} = 0, \quad f_{yy} = 12y^2 = 0$$

$$f(x, y) = 2x^4 + y^4 \geq 0 = f(0, 0) \text{ for any } (x, y) \in \mathbb{R}^2$$

$\Rightarrow (0, 0)$ is a local/global minimum

b. $f_x = -y^2$ $f_y = -2xy \Rightarrow (a, 0)$ for any $a \in \mathbb{R}$ is a critical point.

$$(a, 0): \quad f_{xx} = 0, \quad f_{xy} = -2y = 0, \quad f_{yy} = -2x = -2a$$

$$(a, 0): \quad \begin{cases} a > 0 & f(a, y) = 2 - ay^2 \leq 2 = f(a, 0) & \text{local maximum} \\ a < 0 & f(a, y) = 2 - ay^2 \geq 2 = f(a, 0) & \text{local minimum} \\ a = 0 & f(0, 0) = 2 & \text{saddle point.} \end{cases}$$

Lagrange's Multiplier

Procedure. Finding Absolute Maximum/Minimum Values on Closed, Bounded Sets,

f is defined on a closed bounded set R on $\mathbb{R}^2$

1. Determine the values of f at all critical points in R
2. Find the maximum and minimum values of f on the boundary of R
3. The greatest/least function value found in step 1 and 2 is the absolute

maximum/minimum value of f on R .

The problem becomes

$$\text{Maximize/Minimize } f(x,y) \quad \text{st. } g(x,y) = c$$

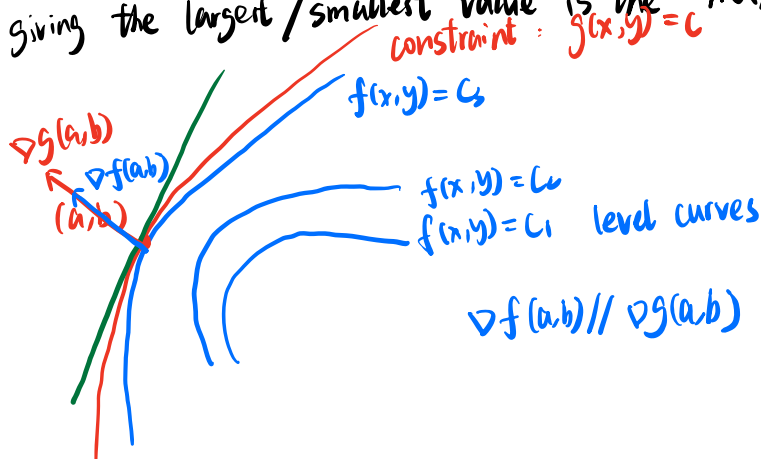
Step 1. Solve the system of equations

$$\begin{cases} \nabla f(x,y) = \lambda \nabla g(x,y) \\ g(x,y) = c \end{cases}$$

Here the unknowns are x, y, λ . ↙ Lagrange's multiplier

Step 2. The solutions of (x,y) are candidates of the maximum or minimum points. We call them boundary critical points.

Step 3. Evaluate $f(x,y)$ at each boundary critical points. The point giving the largest/smallest value is the maximum/minimum on the boundary.



Boundary $g(x,y) = c$ parameterized by $\vec{r}(t) = \langle x(t), y(t) \rangle$.

Maximize/minimize $f(x(t), y(t))$

$$\begin{aligned} 0 = \frac{df}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle \cdot \left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle \\ &= \nabla f \cdot \vec{r}'(t) \end{aligned}$$

$$\begin{cases} \nabla f \perp \vec{r}'(t) \\ \nabla g \perp \vec{r}'(t) \end{cases} \Rightarrow \nabla f \parallel \nabla g \quad \text{at the maximum/minimum point.}$$

Example. $f(x,y) = x^2 + y^2 + 2x + 2y$, find the maximum and minimum values of f when (x,y) is restricted on the constraint $x^2 + y^2 = 1$.

Sol. $g(x,y) = x^2 + y^2$

$$\nabla f = \langle 2x+2, 2y+2 \rangle \quad \nabla g = \langle 2x, 2y \rangle$$

$$\begin{cases} 2x+2 = 2\lambda x \\ 2y+2 = 2\lambda y \\ x^2 + y^2 = 1 \end{cases} \Rightarrow (x,y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = 1 + 2\sqrt{2} \quad \text{maximum}$$

$$f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = 1 - 2\sqrt{2} \quad \text{minimum.}$$